



Creative T3150

Traditional looking 2.1 speakers with bluetooth compatibility, rumored to have an Image Focusing Plate. Assumed to be priced at around USD 100. <http://digit.in/16RdNeE>

Stanford adopts EdX

Stanford University will integrate EdX, a not-for-profit platform founded by Harvard and MIT, with its online classes from June 1st

Browser Engine 101

So, what's the deal with Google announcing a new browser engine earlier this month? A lot apparently if you've been following the speculation by analysts about the motives of Google. Google announced that it would stop using WebKit – the browser engine that's behind displaying web pages in Chrome – and will now be forking WebKit to create its own new Blink browser engine

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Let's start from what we know. You know what a browser is – you browse the Internet with it. And a browser engine helps the browser that lets you browse. Earlier, the web browsers were singular. The first browser, the World-WideWeb, which was later renamed Nexus to avoid confusion, would only follow a few steps of processing text and converting it to a visual representation. The browsers we see now are modular. Here are most of the browsers used today: Chrome, Safari, Internet Explorer, Firefox and Opera. We can divide the web page into two parts: the top part and the main part. In other words the chrome (no, not Google Chrome) and the content.

The chrome, in all of the browsers, will have the top common area, and the content is the main information on the page. The chrome is the generic layout of the browser, like the chrome on the car (shiny stuff) and the rest of the innards of the car are the content. The address bar, stop and refresh buttons, back and forward buttons, the home button and other generic functions like bookmarking favourites, download manager are common to most browsers. So that's the chrome.

The content is what changes; it has the information. For example, the text, images or then plugins and extensions. And the browser engine is what does most of the work in gathering the information of the content. The browser engine is embedded in the browser and it fetches the docs, images and other content that load on the web page. The host application in the browser is responsible for the peripheral, encapsulating chrome and the browser engine fetches the core information on the page.

How the engine works

The engine in general has two things to contend with. The HTML (latest version HTML5, but will probably take till 2014 for the standard to be stable) and the CSS (latest version CSS3).



It uses an HTML parser and a CSS parser after fetching the HTML and CSS content. A parser translates a document to a structure that will make sense to the engine. The translated HTML and CSS document is then used in the next step to construct a render tree. A render tree or a render object is a tree of visual elements in the order in which the elements will be displayed. Each renderer will specify a rectangular area with geometric information like width and height. In the next step the browser engine performs the layout of the render tree. In the layout step the position and size of the nodes or the renderers are specified. The last step is painting. The engine traverses the render tree and paints to display the contents of the page starting with the background colour, images and then outline. Painting used to be done by the CPU but now more of it is given to the graphics card. Finally the engine downloads and runs a Java Script if the web page has one. The new Java Script engines store information about the code so that it needs to be coded only once, which makes the browser faster.

So the browser engine handles the user interface as it works on fetching the content

and layout of the requested page. The browser engines used in some of the most popular browsers are: *Gecko* in Firefox; *WebKit* in Chrome and Safari (both of which are open source engines); *Presto* in Opera; and *Trident* in Internet Explorer (which are not open source).

Blink to see the future

Coming back to where we started, why is the Google news worth discussing? Google will be forking WebKit for its new engine Blink. WebKit is the most widely used engine through Safari and Chrome that accounts for about 45 per cent in terms of usage. It might be lobbying to make the technology it develops for Blink to be used as a web standard. Google says the motivation behind this step is to increase the "collective pace of innovation" but it could be something just as obvious as to snub Apple (Safari is Apple's browser). But Google has also started trimming the code from WebKit in Blink, and is said to have trimmed it by almost 4.5 million lines. Now, if Google's Blink is pushed as a web standard, that'll be something most of us will be keeping a close eye on. 